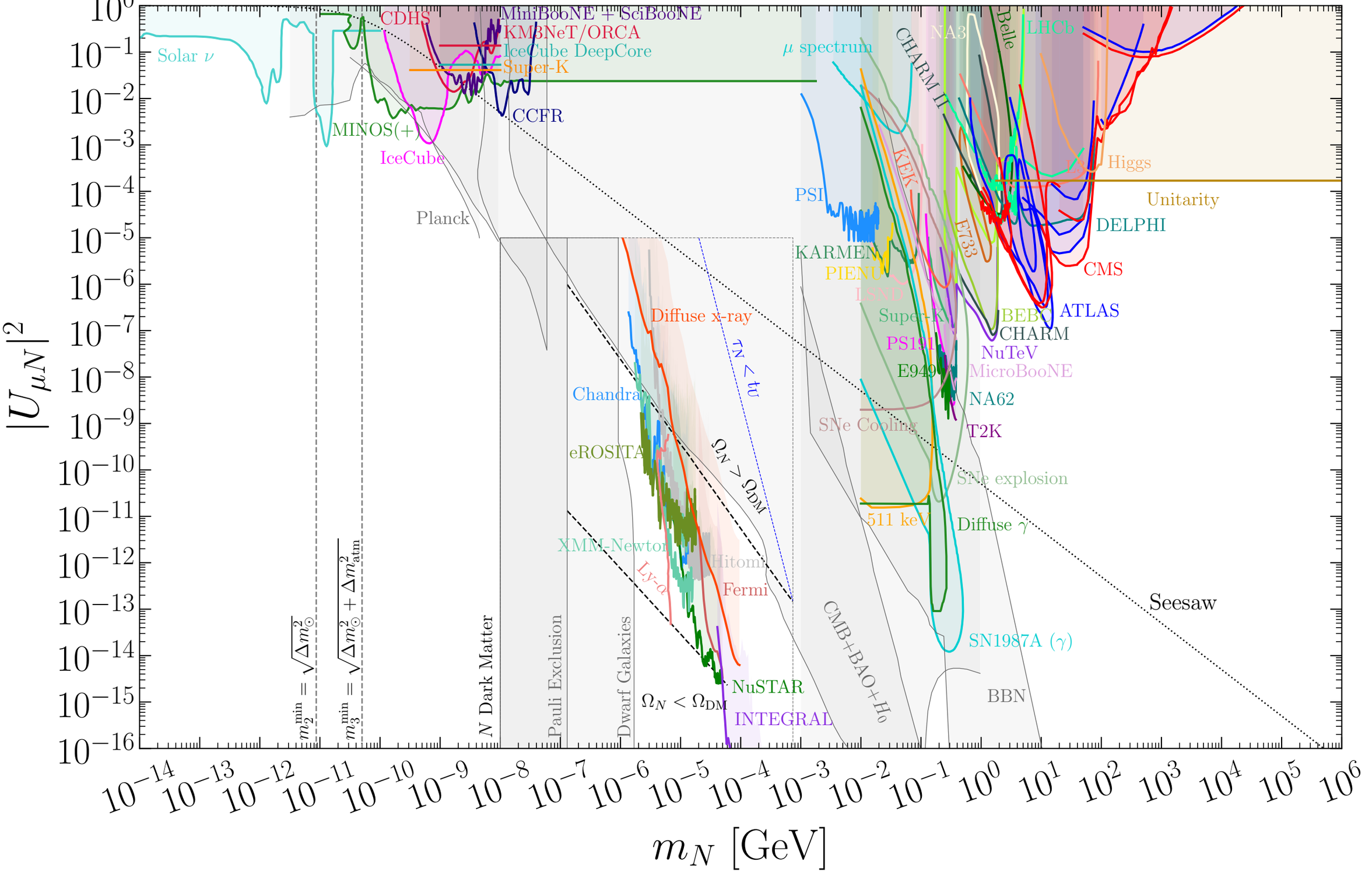

$|U_{\mu N}|^2$
$m_N$ [GeV]
Solar $\nu$
CDHS
MiniBooNE + SciBooNE
KM3NeT/ORCA
IceCube DeepCore
Super-K
CCFR
MINOS(+)
IceCube
Planck
$\mu$ spectrum
CHARM II
NA3
Belle
LHCb
KEK
PSI
E733
Higgs
Unitarity
DELPHI
CMS
ATLAS
KARMEN
PIENU
LSND
Super-K
BEBC
CHARM
PS191
NuTeV
E949
MicroBooNE
NA62
T2K
Diffuse x-ray
$\tau_N < t_U$
Chandra
eROSITA
SNe Cooling
SNe explosion
$\Omega_N > \Omega_{\rm DM}$
511 keV
Diffuse $\gamma$
XMM-Newton
Hitomi
Ly-$\alpha$
Fermi
Seesaw
SN1987A ($\gamma$)
$m_2^{\rm min} = \sqrt{\Delta m^2_\odot}$
$m_3^{\rm min} = \sqrt{\Delta m^2_\odot + \Delta m^2_{\rm atm}}$
$N$ Dark Matter
Pauli Exclusion
Dwarf Galaxies
CMB+BAO+$H_0$
BBN
$\Omega_N < \Omega_{\rm DM}$
NuSTAR
INTEGRAL